

Module 8 Social Network Project:

The application of network analysis to sports data has been increasing in recent years, providing a new view through which player relationships and club structures can be examined. Instead of evaluating players solely on individual performance metrics, social network analytics can show us how players are positioned within the broader ecosystem of connections. In the context of professional soccer, these connections reveal patterns that traditional statistics alone cannot capture. We use the FIFA 22 Complete Player Dataset from Kaggle, which contains detailed attributes for over 19,000 players from across the globe across different soccer leagues. For the analysis, we used the top 500 players by overall rating to keep the network manageable and ensure that the most relevant and well-known players are represented. The network was constructed by treating players as nodes and drawing edges between any two players who share the same club. We ask the question: who are the most influential players in the soccer network? In order to come to our answer, we use three centrality measures: degree centrality, betweenness centrality, and PageRank. In addition, we also use community detection to help identify natural groupings within the network.

Data and Network Preparation

The dataset contains a total of 19,239 entries with many columns describing the attributes of players, from overall rating to the club they play at. After filtering to the top 500 players by overall rating, the dataset represented 107 unique clubs across 22 leagues and 58 nationalities. The network was created using the NetworkX library, where each player was added as a node carrying attributes that include club, nationality, overall rating, market value, and league. Edges were then drawn between every pair of players who shared the same club, creating a co-club connection network. Below are the statistics of the network:

Nodes (players): 499

Edges (co-club): 2283

Network density: 0.0184

Is connected: False

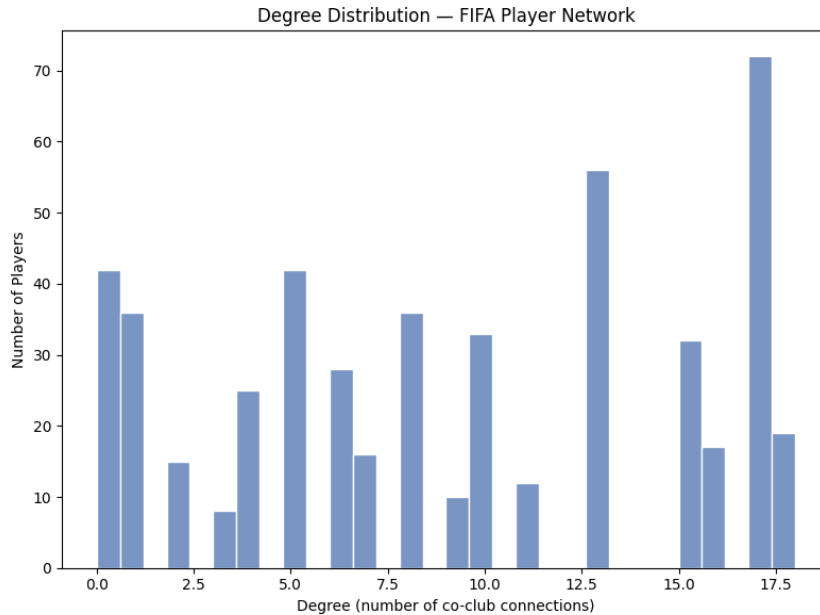
Degree — Mean: 9.15

Degree — Max: 18

Degree — Min: 0

Degree — Std: 5.92

As we investigate the statistics, we can see that the network contained 499 nodes and 2,283 edges. The network density of 0.0184 means that about 1.8% of all possible connections between players actually exist. This is expected given that edges are restricted to shared club membership. The network not being fully connected (499 nodes instead of 500) is due to one player having a degree of zero, meaning that player was the sole representative of their club in the top 500 list, leaving them with no connections to any other player in the dataset. Now let us take a look at the degree distribution.

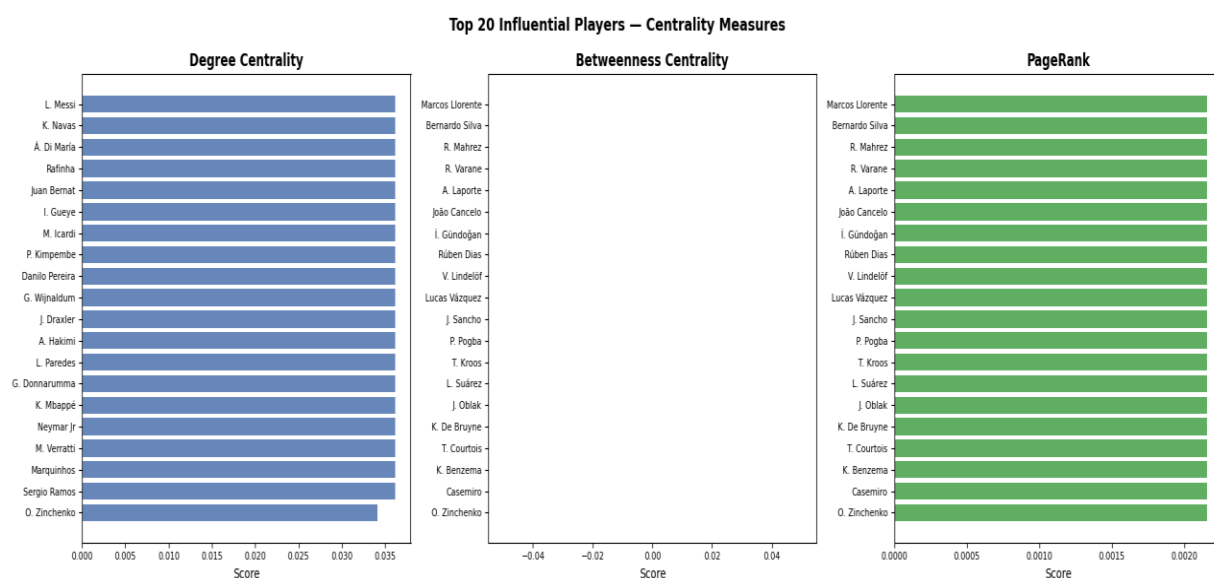


As we look at the visualization, we can see that the chart does not follow a smooth bell curve as is typical in many social networks. This visualization shows an irregular pattern with two notable peaks: one at the low end representing players from clubs with only a few top-rated players in the dataset, and another at the high end representing players from elite clubs such as Real Madrid, PSG, and Manchester City, whose entire starting lineups are represented in the top 500 list. This uneven distribution reflects the real-world financial hierarchy in professional soccer, where only a small number of wealthy clubs have an elite pool of players, while other clubs have a mix of one or two superstars alongside well-known squad players.

Centrality Measures

Three centrality measures were calculated to identify the most influential players in the network. Each measure captures a different element of influence, and when combined, they provide a clearer picture than any single metric alone. Degree centrality measures how many other players a given player is directly connected to. In this network, a high degree score suggests that a player belongs to a club with many other highly rated players in the dataset.

Betweenness centrality measures how often a player sits on the shortest path between two other players, identifying those who act as bridges between different parts of the network. PageRank measures influence based on the quality of connections rather than the number of connections, so a player connected to other highly connected players receives a higher PageRank score. To keep consistency and make the charts easier to read, we used a top 20 list of influential players across all three metrics.



As we investigate the charts, there are a few things to consider. For degree centrality, we can see that the chart is dominated mostly by PSG players. In FIFA 22, PSG at the time had a multitude of highly notable players, from the likes of Neymar, Mbappé, Sergio Ramos, and Messi, who leads the chart at the top. This makes sense as degree centrality, specifically in this network, measures the number of elite teammates a player has. With names like those mentioned, it is no wonder the chart was dominated by PSG. The betweenness centrality results show no meaningful values. This is a direct consequence of how the network was constructed, because edges only exist between players who share a club, no player can genuinely bridge two separate club communities. A dataset that included a player's full club history would have been

better suited for capturing this measure. Finally, looking at PageRank, we can see that the results differ from degree centrality, with Marcos Llorente and Bernardo Silva leading the way. This is because PageRank rewards players whose connections are themselves highly connected. Players at clubs like Atlético de Madrid and Manchester City receive elevated scores due to their multiple interconnections with highly rated players, despite having a lower raw degree count compared to the PSG group.

Community Detection

Community detection was performed to identify natural groupings within the network by maximizing the density of connections within communities relative to connections between them. Below is the terminal output:

Communities found: 106

Community 0: 19 players | Dominant league: French Ligue 1

Community 1: 18 players | Dominant league: English Premier League

Community 2: 18 players | Dominant league: Spain Primera Division

Community 3: 18 players | Dominant league: Spain Primera Division

Community 4: 18 players | Dominant league: English Premier League

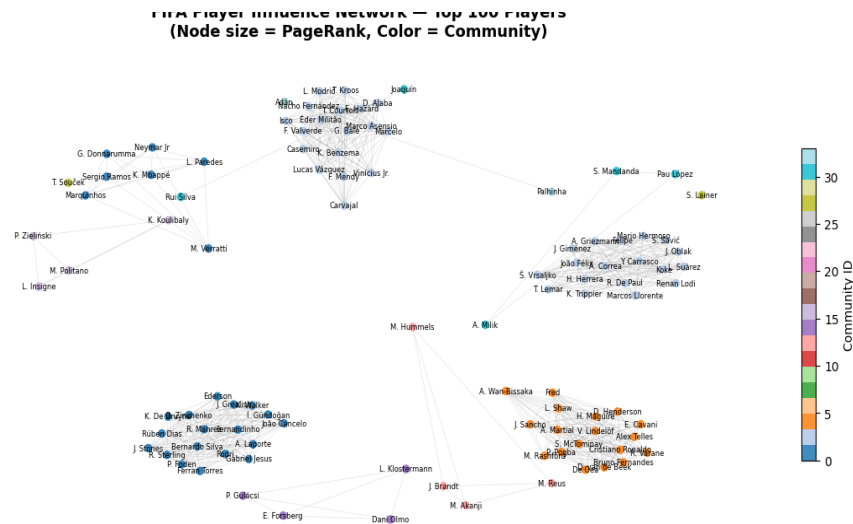
Total players (nodes): 499

Total connections (edges): 2283

Network density: 0.0184

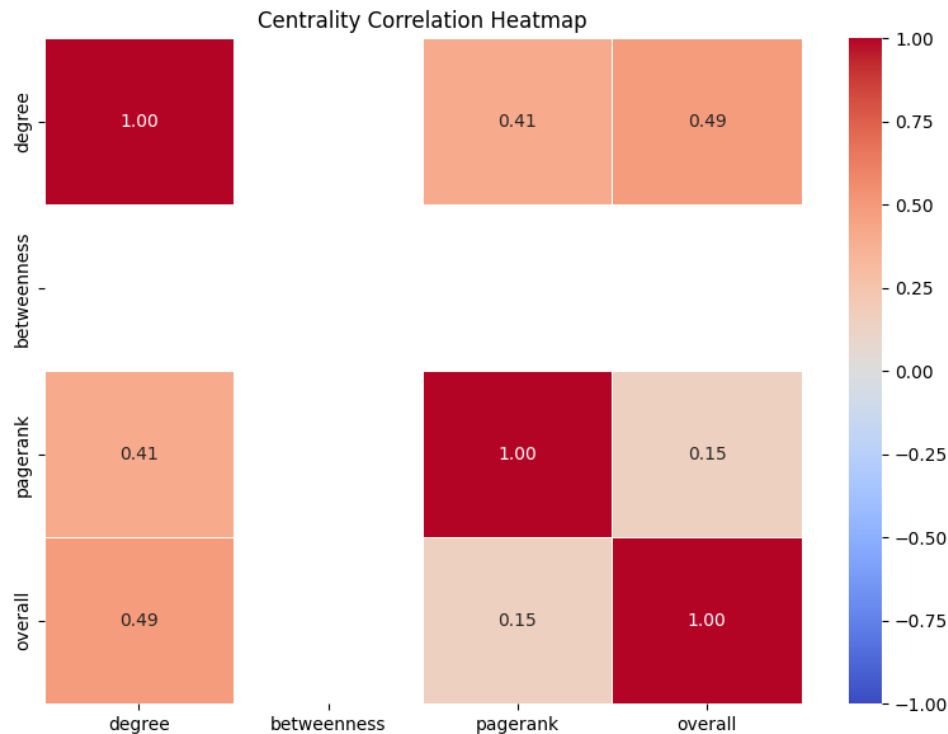
Communities detected: 106

Given this information, we also created a network graph to give us a visualization:



As we can see, 106 communities were detected from 107 different clubs, primarily dominated by the top five European leagues. The graph shows the top 100 players and their connections, where each cluster of nodes represents a single club. Node size reflects PageRank score and the colors represent the respective community. The clear distinction between clusters confirms that club communities are well defined and self-contained.

One last visual to look at is the heatmap, where we examine the relationships between the three centrality measurements and overall player ratings.



As we look at the heatmap, we can see a moderate positive correlation of 0.49 between degree centrality and overall rating. This suggests that highly rated players are often at clubs with other highly rated players, which is about what you would expect in real-world soccer. Though it is only moderate, because while a high rating often means high-rated teammates, that is not always a guarantee, clubs go through transitions, and a highly rated team one year may drop to just below the top tier the next, depending on how you measure a team's performance. We already discussed betweenness, so it makes sense why that is blank here as well. Looking at PageRank, the correlation with overall rating is weaker at 0.15, meaning PageRank captures a different dimension of influence that is less tied to individual player quality.

This analysis was genuinely enjoyable to create and investigate. Taking a trip down memory lane and revisiting that year's ratings, seeing who ranked highly depending on the metric, recalling the immense squad PSG had at the time, and noticing how many well-connected players Manchester City had as well. It is important to note that this dataset has its limitations

because it captures only a single year and is based on the statistics of a video game. There is no transfer history, no outside factors such as nationality or brand associations, or anything of that nature that could have enriched the analysis further. Despite that, it does allow us to grasp the tip of the iceberg of what social network analysis can offer and see how this type of analysis can be applied to real-world problems in soccer and beyond.